

HART ISLAND, NS, CanadaWMO: **714190**Lat: **45.344N**Lon: **60.990W**Elev: **27**StdP: **14.68**Time Zone: **-4.00 (NAA)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	6.8	10.8	-3.2	4.6	9.5	1.0	5.8	12.6	43.9	24.3	41.2	25.5	19.3	290	0.811

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.4	74.4	66.2	71.9	65.3	70.0	64.7	69.1	71.6	67.7	70.1	66.3	68.7	12.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
68.2	103.7	70.2	66.8	98.7	69.0	65.4	93.8	67.7	33.2	71.7	32.1	70.0	31.0	68.6	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
37.1	33.6	29.8	0.5	80.6	4.7	3.4	-2.8	83.0	-5.6	85.0	-8.2	86.9	-11.6	89.3
			-0.5	71.5	4.6	2.4	-3.9	73.2	-6.6	74.6	-9.2	75.9	-12.5	77.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	44.7	27.0	26.1	30.3	37.2	45.2	53.8	62.5	65.5	60.6	51.1	42.1	33.4	(6)
(7)		DBStd	14.67	8.19	7.65	6.14	4.54	4.79	5.04	3.95	3.07	4.68	5.48	6.53	7.26	(7)
(8)		HDD50	3373	712	669	612	383	164	19	0	0	0	53	246	515	(8)
(9)		HDD65	7492	1177	1089	1077	833	613	336	97	30	144	430	686	980	(9)
(10)		CDD50	1438	0	0	0	0	17	134	388	482	319	88	10	0	(10)
(11)		CDD65	79	0	0	0	0	0	0	19	47	13	0	0	0	(11)
(12)		CDH74	92	0	0	0	0	0	2	33	42	15	0	0	0	(12)
(13)		CDH80	6	0	0	0	0	0	0	2	2	2	0	0	0	(13)
(14)	Wind	WSAvg	15.2	18.1	17.5	16.5	15.0	13.4	12.7	12.5	12.4	14.0	15.5	16.6	18.1	(14)
(15)	Precipitation	PrecAvg	59.1	5.8	5.2	4.6	4.4	4.0	4.1	3.5	3.9	4.7	6.0	6.0	6.8	(15)
(16)		PrecMax	72.9	8.9	7.2	8.7	7.4	7.0	6.8	6.7	9.0	12.1	10.5	9.5	14.0	(16)
(17)		PrecMin	45.3	2.8	1.9	1.9	2.3	1.9	2.0	1.5	1.7	1.7	2.4	3.8	3.8	(17)
(18)		PrecStd	6.0	1.5	1.5	1.7	1.3	1.4	1.5	1.2	1.3	2.4	2.2	1.6	2.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.5	45.9	47.4	53.4	66.0	71.9	77.9	78.0	75.7	66.9	59.7	52.8	(19)
(20)			MCWB	48.0	45.2	43.0	46.8	53.8	60.5	68.1	67.3	67.6	63.0	57.1	51.8	(20)
(21)		2%	DB	44.8	42.0	43.4	49.4	59.5	67.4	73.7	74.7	71.6	63.8	56.5	49.4	(21)
(22)			MCWB	44.0	41.0	41.3	44.5	51.7	58.4	65.6	66.6	66.5	61.1	54.6	48.0	(22)
(23)		5%	DB	41.3	38.7	40.5	46.3	55.4	64.3	71.0	72.4	69.0	61.4	54.1	46.6	(23)
(24)			MCWB	40.2	37.6	38.4	43.1	48.9	57.2	64.3	65.8	65.2	58.6	52.5	45.3	(24)
(25)		10%	DB	38.1	35.9	37.8	43.9	52.2	61.6	68.9	70.5	67.1	59.3	51.8	43.5	(25)
(26)			MCWB	36.6	34.7	35.6	40.9	47.4	56.2	63.5	65.2	63.5	56.4	50.0	42.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.3	45.5	45.0	50.1	56.6	64.5	70.0	71.2	70.6	64.8	57.9	52.1	(27)
(28)			MDCB	48.7	46.1	46.1	51.6	62.9	68.0	73.6	73.8	73.2	65.5	58.9	52.4	(28)
(29)		2%	WB	44.1	41.7	42.1	46.3	52.9	61.5	68.0	69.5	68.3	62.0	55.1	48.3	(29)
(30)			MDCB	44.6	42.2	43.1	48.0	57.4	64.1	70.9	71.9	70.5	63.4	55.9	49.1	(30)
(31)		5%	WB	40.4	38.0	39.0	43.9	50.3	59.3	66.5	68.2	66.3	59.3	52.6	45.4	(31)
(32)			MDCB	41.1	38.7	40.1	45.8	53.9	62.1	69.0	70.5	68.3	60.9	53.5	46.3	(32)
(33)		10%	WB	36.9	35.0	36.3	41.5	48.3	57.2	64.9	67.0	64.6	57.0	50.1	42.1	(33)
(34)			MDCB	38.0	35.8	37.6	43.5	51.3	60.5	67.3	69.3	66.4	58.8	51.4	43.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.7	10.7	9.3	9.4	11.1	11.4	10.7	9.4	9.3	8.9	9.1	9.7	(35)
(36)			MCDBR	15.4	14.4	12.9	13.4	17.8	16.6	14.4	12.2	11.1	10.9	11.8	13.4	(36)
(37)			MCWBR	16.3	14.6	11.8	10.3	10.6	9.7	8.1	6.9	7.8	10.1	11.9	14.1	(37)
(38)		5% WB	MCDBR	15.4	14.1	12.7	12.5	15.4	13.9	12.0	9.8	10.1	10.6	11.3	13.4	(38)
(39)			MCWBR	16.6	14.8	12.6	11.0	10.6	10.1	8.0	7.0	8.3	10.8	12.4	14.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.303	0.297	0.323	0.357	0.374	0.393	0.422	0.401	0.361	0.339	0.322	0.301	(40)	
(41)		taud	2.459	2.498	2.462	2.425	2.429	2.415	2.371	2.421	2.508	2.545	2.524	2.494	(41)	
(42)		Ebn at Noon	257	284	289	287	284	278	268	270	273	265	248	246	(42)	
(43)		Edh at Noon	22	25	30	34	36	36	37	34	29	24	21	19	(43)	
(44)	All-Sky Solar Radiation	RadAvg	389	672	1077	1340	1587	1694	1679	1534	1232	782	438	297	(44)	
(45)		RadStd	28	62	67	103	163	151	116	104	84	69	51	21	(45)	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	+1.40	N/A	N/A	N/A	N/A	N/A	+45
(47)	Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.77	N/A	N/A	N/A	N/A

Nomenclature: See separate page